

Qingzhao Zhang

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EDUCATION

University of Michigan <i>Ph.D. in Computer Science and Engineering</i>	<i>Sep 2019 – May 2025</i>
University of Michigan <i>M.S. in Computer Science and Engineering</i>	<i>Sep 2019 – Dec 2023</i>
Shanghai Jiao Tong University <i>B.E. in Computer Engineering</i>	<i>Sep 2015 – May 2019</i>

RESEARCH INTERESTS

System security (e.g., cyber-physical systems, autonomous driving), software security (e.g., program analysis, formal verification), AI security (e.g., adversarial attacks, robustness).

WORK EXPERIENCE

Assistant Professor University of Arizona, ECE • Lead research group working on computer security and systems. We advance the security, resilience, and trustworthiness of emerging autonomous systems (e.g., self-driving vehicles, drones, and robots) through cross-layer design and analysis.	<i>Aug 2025 - Present</i>
Research Assistant/Research Fellow University of Michigan, EECS <i>Supervisor: Prof. Z. Morley Mao</i> • Research on software/system for cyber-physical system safety — <i>AVChecker</i> , the first traffic rule compliance checker on autonomous driving software; <i>SmtConf</i> , safety vetting of industrial control system configuration. • Research on adversarial machine learning on cyber-physical systems — Adversarial attack and mitigation on trajectory prediction on autonomous driving; Analyzed data fabrication vulnerability on collaborative perception. • Research on robustness of vehicular network — <i>RAO</i> , collaborative perception with asynchronous sensors. • Research on robust perception algorithms of autonomous driving, and large language model security/efficiency.	<i>May 2020 - Aug 2025</i>
Software Engineer Intern Google • Designed and implemented formal verification solutions to enhance the security properties of an embedded system kernel (based on open-sourced Tock OS), involving modular model checking and theorem proving.	<i>May 2023 - July 2023</i>
Software Engineer Intern Google • Designed and implemented static analysis checks based on Android Lint for Google’s Android tests, which was deployed to assist Google developers to write high-quality unit tests.	<i>May 2022 - July 2022</i>

SELECTED PUBLICATIONS (*: co-primary)

Conference papers

- [COLM’25] Shuowei Jin, Yongji Wu, Haizhong Zheng, **Qingzhao Zhang**, Matthew Lentz, Z Morley Mao, Atul Prakash, Feng Qian, Danyang Zhuo. “Plato: Plan to efficiently decode for large language model inference”, *Conference on Language Modeling 2025*.
- [ICML’25] Shuowei Jin, Xueshen Liu, **Qingzhao Zhang**, Z. Morley Mao. “Compute Or Load KV Cache? Why Not Both?”, *The 42nd International Conference on Machine Learning*.

- [ICLR'25] Minkyong Cho, Yulong Cao, Jiachen Sun, **Qingzhao Zhang**, Marco Pavone, Jeong Joon Park, Heng Yang, Z. Morley Mao. “Cocoon: Robust Multi-Modal Perception with Uncertainty-Aware Sensor Fusion”, *The 13th International Conference on Learning Representations*.
- [Security'24] **Qingzhao Zhang**, Shuwei Jin, Ruiyang Zhu, Jiachen Sun, Xumiao Zhang, Qi Alfred Chen, Z. Morley Mao. “On Data Fabrication in Collaborative Vehicular Perception: Attacks and Countermeasures”, *The 33th USENIX Security Symposium*.
- [ICLR'24] Jiachen Sun, Haizhong Zheng, **Qingzhao Zhang**, Atul Prakash, Z. Morley Mao, Chaowei Xiao. “CALICO: Self-Supervised Camera-LiDAR Contrastive Pre-training for BEV Perception”, *The 12th International Conference on Learning Representations*.
- [Mobicom'23] **Qingzhao Zhang***, Xumiao Zhang*, Ruiyang Zhu*, Fan Bai, Mohammad Naserian, Z. Morley Mao. “Robust Real-time Multi-vehicle Collaboration on Asynchronous Sensors”, *The 29th International Conference on Mobile Computing and Networking*.
- [CVPR'22] **Qingzhao Zhang**, Shengtuo Hu, Jiachen Sun, Qi Alfred Chen, Z. Morley Mao. “On Adversarial Robustness of Trajectory Prediction for Autonomous Vehicles”, *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- [RAID'22] **Qingzhao Zhang**, Xiao Zhu, Mu Zhang, Z. Morley Mao. “Automated Runtime Mitigation for Timing-based Safety Hazards in Industrial Controllers”, *The 25th International Symposium on Research in Attacks, Intrusions and Defenses*.
- [AsiaCCS'22] Shengtuo Hu, **Qingzhao Zhang**, André Weimerskirch, Z. Morley Mao. “GateKeeper: A Gateway-based Broadcast Authentication Protocol for the In-vehicle Ethernet”, *Proceedings of the ACM on Asia Conference on Computer and Communications Security*.
- [IV'22] Ze Zhang, Sanjay Sri Vallabh Singapuram, **Qingzhao Zhang**, David Ke Hong, Brandon Nguyen, Z. Morley Mao, Scott Mahlke, Qi Alfred Chen. “AVMaestro: A Centralized Policy Enforcement Framework for Safe Autonomous-driving Environments”, *IEEE Intelligent Vehicles Symposium*.
- [SIGMETRICS'21] **Qingzhao Zhang**, David Ke Hong, Ze Zhang, Qi Alfred Chen, Scott Mahlke, Z. Morley Mao. “A Systematic Framework for Checking Driving Rule Compliance in Autonomous Vehicle Software”, *Proceedings of the ACM on Measurement and Analysis of Computing Systems*.
- [SANER'20] **Qingzhao Zhang***, Yizhuo Wang*, Juanru Li, Siqi Ma. “EthPloit: From Fuzzing to Efficient Exploit Generation against Smart Contracts”, *IEEE 27th International Conference on Software Analysis, Evolution and Reengineering*.
- [FC'20] Tsz Hon Yuen, Shi-feng Sun, Joseph K Liu, Man Ho Au, Muhammed F Esgin, **Qingzhao Zhang**, Dawu Gu. “Ringct 3.0 for blockchain confidential transaction: Shorter size and stronger security”, *Financial Cryptography and Data Security: 24th International Conference*.

Journal papers

- [TDSC] Yanxue Jia, Shi-Feng Sun, Yuncong Zhang, **Qingzhao Zhang**, Ning Ding, Zhiqiang Liu, Joseph K Liu, Dawu Gu. “PBT: A New Privacy-Preserving Payment Protocol for Blockchain Transactions”, *IEEE Transactions on Dependable and Secure Computing*.

Workshop and Work-in-Progress papers

- [VehicleSec'25] Jiarui Li, Joseph Brewington, **Qingzhao Zhang**, Z. Morley Mao. “WIP: Hijacking Attacks on UAV Follow-Me Systems in Realistic Scenarios”, *3rd USENIX Symposium on Vehicle Security and Privacy*.
- [LAMPS'25] **Qingzhao Zhang**, Ziyang Xiong, Z. Morley Mao. “Safeguard is a Double-edged Sword: Denial-of-service Attack on Large Language Models”, *2nd ACM Workshop on Large AI Systems and Models with Privacy and Safety Analysis*.
- [CPSIoTSec'24] **Qingzhao Zhang**, Z. Morley Mao. “Stealthy Data Fabrication in Collaborative Vehicular Perception”, *The 6th Workshop on CPS and IoT Security*.

- **[SRML'22]** Jiachen Sun, **Qingzhao Zhang**, Bhavya Kailkhura, Zhiding Yu, Chaowei Xiao, Z Morley Mao. “ModelNet40-C: A Robustness Benchmark for 3D Point Cloud Recognition Under Corruption”, *ICLR 2022 Workshop on Socially Responsible Machine Learning*.
- **[FEAST'20]** Ze Zhang, **Qingzhao Zhang**, Brandon Nguyen, Sanjay Sri Vallabh Singapuram, Z Morley Mao, Scott Mahlke. “Automatic Feature Isolation in Network Protocol Software Implementations”, *ACM Workshop on Forming an Ecosystem Around Software Transformation*.

Preprints & In submission

- **Qingzhao Zhang***, Shaocheng Luo*, Z. Morley Mao, Miroslav Pajic, Michael K. Reiter. “SoK: How Sensor Attacks Disrupt Autonomous Vehicles: An End-to-end Analysis, Challenges, and Missed Threats”.
- Fangzhou Wu, **Qingzhao Zhang**, Ati Priya Bajaj, Tiffany Bao, Ning Zhang, Ruoyu Wang, Chaowei Xiao. “Exploring the Limits of ChatGPT in Software Security Applications”.
- Spencer Hallyburton, **Qingzhao Zhang**, Z. Morley Mao, Miroslav Pajic. “Partial-Information, Longitudinal Cyber Attacks on LiDAR in Autonomous Vehicles”.
- Jiachen Sun, **Qingzhao Zhang**, Bhavya Kailkhura, Zhiding Yu, Chaowei Xiao, Z. Morley Mao. “Benchmarking Robustness of 3d Point Cloud Recognition Against Common Corruptions”.
- Anrin Chakraborti*, **Qingzhao Zhang***, Jingjia Peng, Z. Morley Mao, Michael K. Reiter. “StreetCred: A Privacy-Preserving Reputation System for Connected Autonomous Vehicles”.
- Ruiyang Zhu, **Qingzhao Zhang**, Xumiao Zhang, Fan Bai, Mohammad Naserian, Z. Morley Mao. “RAO++: Realistic Real-time Multi-vehicle Collaboration on Asynchronous Sensors”.

TEACHING AND MENTORSHIP

PhD Students, University of Arizona

- Yutong Liu, started in Fall 2025.

Supervisor of Undergraduate Research, University of Michigan

2022

- Multidisciplinary Design Program (MDP) at University of Michigan. Lead cybersecurity research and supervised a team of four students.

Research Mentorship, University of Michigan

2020-2024

- Research on program analysis, autonomous driving, adversarial machine learning.
- Mentees: Charles Ziegenbein Jr., Kevin Zhang, Andrew Wei, Xingyu Wang, Jingjia Peng, Ziyang Xiong, Runting Zhang.

SERVICES

- Conference Committee Member: CCS 2025, CCS 2026
- Conference Reviewer: CVPR 2026, ICLR 2025, ICRA 2025, ACM MM 2024, IV 2024, IV 2023
- Journal Reviewer: TDSC, IoT-J, Sensors, TIFS, ITSM
- Workshop Committee Member: RICSS 2024, RICSS 2025
- Artifact Reviewer: Usenix Security 2022
- Pre-review Task Force: NSDI 2025

TALKS

• Enhancing Security, Safety, and Reliability of Cyber-physical Systems	
<i>Presentation at Midwest Security Workshop</i>	<i>11/16/2024</i>
• Stealthy Data Fabrication in Collaborative Vehicular Perception	
<i>Presentation at 6th CPSIoTSec Workshop, co-located with CCS 2024</i>	<i>10/18/2024</i>
<i>Poster at Midwest Security Workshop</i>	<i>11/16/2024</i>
• On data fabrication in collaborative vehicular perception: attacks and countermeasures	
<i>Poster at Athena institute (NSF AI Institute) annual showcase (best poster award)</i>	<i>5/3/2023</i>
<i>Presentation at USENIX Security 2024</i>	<i>8/16/2024</i>
• Automated runtime mitigation for timing-based safety hazards in industrial controllers	
<i>Virtual presentation at RAID 2022</i>	<i>10/27/2022</i>
• On adversarial robustness of trajectory prediction for autonomous vehicles	
<i>Poster at Athena institute (NSF AI Institute) annual showcase</i>	<i>8/18/2022</i>
<i>Poster at CVPR 2022</i>	<i>6/23/2022</i>
• Robustness of applications for autonomous vehicles	
<i>Presentation at Workshop on Future Automotive Research Datasets</i>	<i>11/19/2021</i>
• A systematic framework for checking driving rule compliance in autonomous vehicle software	
<i>Virtual presentation at SIGMETRICS 2021</i>	<i>6/16/2021</i>

HONORS AND AWARDS

• Student Travel Grant, Usenix Security 2024, SIGMETRICS 2021, CCS 2021	<i>2019 – 2024</i>
• Rackham Student Fellowship, University of Michigan	<i>2019</i>
• Academic Excellence Scholarship of SJTU (top 5%)	<i>2016, 2017, 2018</i>